

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1516
---------------------	---	---------------------	---------

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SRAVAN Y

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Problem:

A retail company needs real-time inventory tracking, demand forecasting, and personalized customer recommendations.

Solution:

- Collect sales, inventory, website, and customer data.
- Use **Apache Kafka** for real-time data streaming.
- Store large datasets using **Hadoop HDFS** or cloud storage.
- Use **Apache Spark** for large-scale data processing.
- Use **Spark Streaming** for real-time inventory updates.
- Use **Time-Series/ARIMA or LSTM models** for demand forecasting.
- Use **Collaborative Filtering** for personalized recommendations.
- Use **Power BI/Tableau** for business dashboards.

Data**Flow:**

Stores + Website → Kafka → Data Lake → Spark → ML Models → Dashboard/Recommendations

Application:

Used by **Amazon, Walmart, supermarkets, and e-commerce platforms** for inventory and customer personalization.

2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Problem:

A social media platform needs to detect fake accounts and misinformation from billions of interactions.

Solution:

- Collect posts, comments, likes, shares, and account activity.
- Use **Apache Kafka** for real-time event streaming.
- Use **Apache Spark Streaming** or **Apache Flink** for real-time processing.
- Store historical data in a **Data Lake/Hadoop** environment.
- Use **Machine Learning** to identify suspicious behavior.
- Use **Random Forest/XGBoost** for fake-account classification.
- Use **NLP and Transformer models** to detect misinformation.
- Generate alerts when suspicious activity is detected.

Data**Flow:**

User Activity → Kafka → Stream Processing → ML/NLP Models → Fraud Alert

Application:

Used in **social networking platforms, online communities, and content-sharing applications**.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Problem:

Government departments need to share census, economic, and geospatial data securely while maintaining interoperability and privacy.

Solution:

- Collect data from different government departments.
- Store structured and unstructured data in a **Government Data Lake**.
- Use **ETL/ELT tools** for data integration.

- Use standardized formats such as **JSON, XML, CSV, and GeoJSON**.
- Use APIs for secure data sharing between departments.
- Apply **data governance policies** for ownership, quality, and standards.
- Use **role-based access control (RBAC)**.
- Encrypt data during storage and transmission.
- Use **data masking and anonymization** for sensitive citizen information.
- Maintain audit logs for monitoring access.

Data

Flow:

Government Departments → Data Integration → Data Lake → Governance & Security → Shared Data Platform

Application:

Used for **census management, public services, economic planning, smart cities, and national policy planning**.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Problem: A bank processes millions of transactions and must detect fraudulent transactions with very low latency.

Solution:

- Collect transaction data from ATMs, cards, mobile banking, and online banking.
- Use **Apache Kafka** for real-time transaction streaming.
- Use **Apache Flink/Spark Streaming** for real-time processing.
- Store historical transactions in a **Data Lake/Data Warehouse**.
- Use **Random Forest, XGBoost, and Neural Networks** for fraud classification.
- Use **Anomaly Detection** to identify unusual transaction patterns.
- Use historical customer behavior for model training.
- Generate an immediate alert when a transaction is classified as suspicious.
- Continuously update the ML model using new transaction data.

Data Flow:

Transaction → Kafka → Stream Processing → Fraud ML Model → Risk Score → Alert/Block

Application:

Used in **banks, credit-card companies, digital wallets, UPI, and online payment systems**.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Problem:

Farmers need to analyze soil, weather, crop, drone, and sensor data to improve yield and reduce resource usage.

Solution:

- Collect data from **IoT sensors, drones, satellites, weather stations, and farm equipment**.
- Use **MQTT/Kafka** for collecting real-time sensor data.
- Store large datasets in a **Cloud Data Lake**.
- Use **Apache Spark** for large-scale data processing.
- Use **Machine Learning** for crop yield prediction.
- Use **Regression models** for yield forecasting.
- Use **Classification models** for crop disease detection.
- Use sensor analytics to determine irrigation and fertilizer requirements.
- Use dashboards to provide farmers with real-time recommendations.

Data Flow:

Sensors + Drones + Weather Data → Kafka/MQTT → Cloud Data Lake → Spark → ML Analytics → Farmer Dashboard

Application:

Used in **precision farming, crop monitoring, smart irrigation, disease detection, and yield optimization**.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of	Reasonable awareness	Limited awareness	Poor awareness

		security and operational risks			
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization

Key Problem

Retail companies generate huge amounts of data from stores and websites. If this data is not processed properly, companies may face **stock shortages, excess inventory, poor demand prediction, and irrelevant recommendations**.

Important Data

- Sales transactions
- Product inventory
- Customer purchases
- Website clicks
- Customer searches
- Product views
- Location and store data

Big Data Characteristics

- **Volume:** Millions of transactions and customer records
- **Velocity:** Sales and inventory data arrive continuously
- **Variety:** Transaction, customer, web, and product data

Technologies

- **Kafka** – real-time data streaming
- **HDFS/Cloud Data Lake** – large-scale storage
- **Spark** – data processing
- **Spark Streaming** – real-time processing
- **ARIMA/LSTM** – demand forecasting
- **Collaborative Filtering** – recommendations
- **Power BI/Tableau** – visualization

Expected Output

- Current inventory status
- Future demand prediction
- Low-stock alerts
- Recommended products
- Sales dashboards

Main Benefit

It helps the company **reduce inventory problems, improve customer experience, increase sales, and make faster decisions.**

2. Social Media Fraud Detection

Key Problem

Social media platforms receive billions of interactions. Fake accounts, spam, bots, and misinformation can spread very quickly.

Important Data

- Posts and comments
- Likes and shares
- Login activity
- Account creation information
- User interaction patterns
- IP/device behavior
- Text content

Big Data Characteristics

- **Huge Volume:** Billions of interactions
- **High Velocity:** New posts and activities are generated continuously
- **Variety:** Text, images, videos, user activity, and network information

Technologies

- **Kafka** – real-time event collection
- **Spark Streaming/Flink** – stream processing
- **Hadoop/Data Lake** – historical storage

- **Random Forest/XGBoost** – fraud classification
- **NLP/Transformers** – text analysis
- **Machine Learning** – suspicious behavior detection

Important Detection Features

- Unusual login patterns
- Very high posting frequency
- Large numbers of repeated messages
- Abnormal follower/following behavior
- Suspicious links
- Similar activity across multiple accounts

Expected Output

- Genuine/Fake account classification
- Fraud probability
- Misinformation detection
- Suspicious activity alert

Main Benefit

The system can **detect suspicious activity quickly, reduce fake accounts, and improve platform security.**

3. Government Data Platform

Key Problem

Government departments usually maintain different types of data in different systems. Sharing this information can be difficult because of **different formats, security requirements, and data standards.**

Important Data

- Census data
- Economic data
- Citizen service data
- Geographic data
- Population information
- Public infrastructure data
- Department records

Main Requirements

1. **Data Integration** – combine data from different departments.
2. **Interoperability** – allow different systems to communicate.
3. **Data Quality** – remove duplicate and incorrect information.
4. **Security** – protect sensitive information.
5. **Privacy** – prevent unauthorized access.
6. **Governance** – define how data is managed and used.

Technologies

- **ETL/ELT** – data integration
- **Data Lake** – centralized storage
- **JSON/XML/CSV/GeoJSON** – data formats
- **APIs** – secure data sharing
- **RBAC** – access control
- **Encryption** – data protection
- **Data Masking/Anonymization** – privacy
- **Audit Logs** – track access

Expected Output

A centralized platform where authorized government departments can **find, access, exchange, and analyze required data securely**.

Main Benefit

It improves **government decision-making, data sharing, service delivery, transparency, and resource planning**.

4. Banking Fraud Detection

Key Problem

Banks process millions of transactions every day. Fraud must be identified **within seconds**, because delaying detection can result in financial losses.

Important Data

- Transaction amount
- Transaction location
- Time of transaction

- Customer history
- Device information
- Account activity
- Merchant information
- Previous fraud records

Important Fraud Indicators

For example, if a customer normally makes transactions in India but suddenly a large transaction occurs from another location, the system can consider it suspicious.

Other indicators include:

- Unusually large transaction
- Multiple transactions within a short time
- Unusual location
- New device
- Unusual spending pattern
- Repeated failed transactions

Technologies

- **Kafka** – real-time transaction streaming
- **Flink/Spark Streaming** – real-time processing
- **Data Lake/Warehouse** – historical data
- **Random Forest/XGBoost** – classification
- **Neural Networks** – complex fraud patterns
- **Anomaly Detection** – unusual behavior
- **Machine Learning** – fraud prediction

Expected Output

The system generates a **risk score**.

Example:

Transaction	Risk Score	Decision
Normal transaction	10%	Allow
Slightly unusual	55%	Verify
Highly suspicious	95%	Block/Alert

Main Benefit

It helps banks **reduce fraud losses, protect customers, and make fast decisions with low latency**.

5. Agriculture Smart Farming

Key Problem

Farmers need to make decisions about **irrigation, fertilizers, crop diseases, and harvesting**. These decisions can be improved by analyzing large amounts of agricultural data.

Important Data

- Soil moisture
- Soil temperature
- Soil nutrients
- Weather conditions
- Rainfall
- Crop images
- Satellite images
- Drone images
- Crop growth information
- Historical yield data

Technologies

- **IoT Sensors** – collect field data
- **MQTT/Kafka** – data transmission
- **Cloud Data Lake** – storage
- **Spark** – large-scale processing
- **Machine Learning** – prediction
- **Regression** – yield prediction
- **Classification** – disease detection
- **Dashboards** – farmer decision support

Important Applications

- **Smart Irrigation:** Decide when and how much water is needed.
- **Disease Detection:** Identify possible crop diseases.
- **Yield Prediction:** Estimate future crop production.
- **Fertilizer Recommendation:** Suggest fertilizer based on soil conditions.
- **Weather Analysis:** Help farmers plan agricultural activities.

Expected Output

The farmer can receive information such as:

Soil moisture is low → Irrigation required

Weather conditions favorable → Suitable for farming activity

Crop image shows possible disease → Inspection required

Main Benefit

Smart farming can help **increase crop productivity, reduce water and fertilizer wastage, reduce costs, and support better farming decisions.**